CHAPTER 7

Control Unit: Hardwired & Microprogrammed

Micro-operations · hardwired vs microprogrammed · horizontal vs vertical · control store sizing · microprogram sequencing

QUICK TRICK

Hardwired = **fast, rigid** (logic gates). Microprogrammed = **flexible, slower** (control store ROM). Horizontal microinstruction: **1 bit per control signal**, wide, fast. Vertical: **encoded fields**, narrow, needs decoders. A field of n mutually exclusive signals needs $\lceil \log_2(n+1) \rceil$ bits if "no-op" is allowed.

7.1 What does the control unit do?

● **LEVEL 1 · BEGINNER**

The control unit turns each instruction into a **sequence of control signals** (open this register, select that ALU operation, read memory) at the right time. **Analogy.** A conductor: the instrument (data path) plays, and the conductor decides who plays when.

Instruction fetch as micro-operations:

```
T0 : MAR <- PC
T1 : MDR <- M[MAR] ;  PC <- PC + 1
T2 : IR  <- MDR
T3 : decode, then execute micro-operations specific to the opcode
```

7.2 Hardwired control

● **LEVEL 2 · EXAM**

Control signals are produced by a fixed network of gates, decoders and a step counter from (opcode, flags, timing step). Built with state tables and logic minimisation.

7.3 Microprogrammed control

● **LEVEL 2 · EXAM**

Each instruction is implemented by a **microprogram** (a list of microinstructions) stored in a **control store** (ROM). A **microprogram counter (µPC / CAR)** holds the address of the next microinstruction. The opcode is mapped to the starting address of its routine.

	Hardwired	Microprogrammed
Speed	Faster	Slower (control-store access)
Flexibility	Hard to change	Easy: edit the microcode
Design effort	Complex for big ISAs	Systematic, simpler for CISC
Cost	More logic	Needs control memory
Typical	RISC	CISC

7.4 Microinstruction formats

● LEVEL 2 · EXAM

	Horizontal	Vertical
Encoding	One bit per control signal	Signals grouped into encoded fields
Width	Wide	Narrow
Parallelism	High: many signals in one step	Limited
Speed	Faster (no decoding)	Slower (decoders)
Control store	Larger (wide words)	Smaller

Word width = control field + next-address field (+ condition-select bits if given). **Horizontal example:** 40 control signals + 10-bit next address = **50 bits**. **Vertical example:** groups of 5, 8, 3 and 12 mutually exclusive signals, each with a no-op: $\lceil \log_2 6 \rceil + \lceil \log_2 9 \rceil + \lceil \log_2 4 \rceil + \lceil \log_2 13 \rceil = 3 + 4 + 2 + 4 =$ **13 bits**.

Control store size = (number of microinstructions) × (word width). Address bits = $\lceil \log_2(\text{number of words}) \rceil$. **Example:** 512 words × 32 bits = 16 Kbit = **2 KB**, 9 address bits.

7.5 Sequencing and mapping

● LEVEL 3 · GATE

- **Next-address field:** each microinstruction names the next one (or μPC just increments). Conditional branches use a condition-select field and a branch address.
- **Opcode to microaddress:** if each routine has at most 2^m microinstructions and the opcode is k bits, start address = (opcode followed by m zeros). Example: 6-bit opcode, ≤ 4 microinstructions each → start = opcode × 4, address width **8 bits**.
- **Nanoprogramming** (two-level control store) saves space when many microinstructions repeat.

7.6 Common mistakes

COMMON MISTAKE

- Saying vertical microcode is faster (horizontal is faster; vertical is smaller).
- Forgetting the next-address field in the control word width.
- Using $\lceil \log_2 n \rceil$ instead of $\lceil \log_2(n + 1) \rceil$ when a no-operation code is required.
- Forgetting the common fetch routine when counting total control-store words.

QUICK TRICK

Memory line: "Horizontal = wide and quick. Vertical = narrow and encoded."

GATE QUICK CHECK — Chapter 7

```
hardwired fast/rigid/RISC | microprogrammed flexible/slower/CISC
horizontal: width = #signals (+next addr)    vertical: sum ceil(log2(n_i+1))
control store bits = words x width           address bits = ceil(log2 words)
fetch: MAR<-PC ; MDR<-M[MAR], PC<-PC+1 ; IR<-MDR
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. Which control unit is faster, hardwired or microprogrammed?

Q2. Which is easier to modify?

Q3. What does the micro-program counter hold?

Q4. Which format needs decoders, horizontal or vertical?

● LEVEL 2 · EXAM

Q5. Horizontal microinstruction: 40 control signals and a 10-bit next-address field. Word width?

Q6. Vertical format: four groups of 5, 8, 3 and 12 mutually exclusive signals (no-op allowed). Control bits?

Q7. Control store of 512 words × 32 bits: size in KB and address bits.

Q8. Micro-operations for the instruction fetch.

● **LEVEL 3 · GATE**

Q9. A machine has 64 opcodes, each routine averages 8 microinstructions, plus a 12-word common fetch routine. Minimum control-store address bits?

Q10. 6-bit opcode and every routine at most 4 microinstructions. How is the start address formed and how wide is it?

Q11. Why does horizontal microcode allow more parallelism?

Answers

Q	Solution
1	**Hardwired**.
2	**Microprogrammed** (change the microcode).
3	The **address of the next microinstruction**.
4	**Vertical**.
5	40 + 10 = **50 bits**.
6	3 + 4 + 2 + 4 = **13 bits**.
7	512 × 32 = 16384 bits = **2 KB**; **9 address bits**.
8	MAR ← PC; MDR ← M[MAR] and PC ← PC + 1; IR ← MDR.
9	64 × 8 + 12 = 524 words → $\lceil \log_2 524 \rceil$ = **10 bits**.
10	Start address = opcode followed by 2 zero bits (opcode × 4). Width = 6 + 2 = **8 bits**.
11	Each control signal has its own bit, so any combination of signals can be active in the same step.

Next: Chapter 8 — Instruction Pipelining and Hazards.

AMJESH singhamjesh.github.io